

Rayen Mrad

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PROFESSIONAL EXPERIENCE

FlyRank AI

July 2026 – August 2026

Search-Ranking ML Intern

- Built an end-to-end search-ranking pipeline (feature engineering, training, evaluation, reporting) on 30K anonymized content pages using Python, pandas, and scikit-learn, with client-holdout validation and leakage checks for honest results.
- Benchmarked Logistic Regression, Decision Tree, and Random Forest rankers; selected Random Forest for a 3× precision gain over the rule-based baseline (Precision@50: 0.74 vs. 0.24, ROC AUC 0.75).
- Used feature importance analysis (impression trends, average position, content age) to flag a 54% subset of underperforming content for prioritized review.

Welyne — Partnership with SynapsePulse Inc. (USA)

Oct. 2025 – Jan. 2026

AI Intern

Tunis, Tunisia

- Trained Random Forest and Gradient Boosting models on dual-sensor wearable data (cardiac BPM/HRV, motion speed/acceleration) to classify fatigue across 4 severity levels, reaching 74.17% test accuracy and 75.79% macro recall (5-fold CV: 73.33%).
- Built the full ML pipeline end-to-end and deployed it via FastAPI into a live athlete monitoring system, taking the model from notebook to production.

Get Wireless

June 2025 – July 2025

ML Intern

Tunis, Tunisia

- Built a 6-stage image enhancement pipeline (CLAHE, denoising, Gaussian filtering, sharpening, gamma correction, histogram equalization) applied to 200K+ X-ray and PET images.
- Fine-tuned a CheXpert-pretrained DenseNet-121 for multi-label classification of 14 pulmonary pathologies, then adapted it for tuberculosis detection on a public TB Chest X-ray dataset.
- Trained a segmentation model from scratch on LUNA16 for pulmonary nodule detection and segmentation.

PROJECTS

Multi-Modal Industrial Leak Detection System

Apr. 2026

- Built an Isolation Forest anomaly detector on 1.6M readings across 28 sensors using temporal and rolling-window features.
- Designed a 5-stage pipeline (anomaly detection, severity scoring, cost estimation, CNN acoustic confirmation, forecasting) fusing pressure and acoustic signals.
- Deployed real-time inference (FastAPI + WebSocket) to a React dashboard at ~1Hz across 10 simulated pipelines.

ML-Powered Productivity Predictor

Feb. 2026 – Mar. 2026

- Engineered a stacking ensemble (GBR + Random Forest, Ridge meta-learner) with 5-fold CV and polynomial feature expansion to predict productivity from behavioral data.
- Shipped an authenticated FastAPI inference service with lazy model loading and ensemble-based confidence scoring.
- Built a full-stack platform (React, PostgreSQL, Docker) delivering factor-level productivity insights.

Machine Health Monitoring System

Jan. 2026 – Feb. 2026

- Built an end-to-end predictive maintenance pipeline using Python, scikit-learn, and Isolation Forest to detect anomalies across 7 sensor features.
- Generated 100K synthetic time-series samples with 6 failure categories and optimized detection thresholds using validation-based F1 tuning.
- Deployed real-time inference with ESP32, C++, MQTT, and Flask, achieving reported 1.2 ms CPU inference latency.

EDUCATION

National Institute of Applied Sciences and Technology (INSAT)

Sep. 2022 – Present

Industrial IT and Automation Engineering Degree

INSAT, Tunis

- **Coursework:** Embedded Systems, Automation, Electronics, Robotics.

SKILLS

Languages: Python, C/C++, Java, JavaScript/TypeScript, SQL

ML/DL: scikit-learn, pandas, NumPy, CNNs, DenseNet-121, Transfer Learning, Ensemble Methods (Random Forest, Gradient Boosting, Isolation Forest, Stacking)

Techniques: Feature Engineering, Model Evaluation, Anomaly Detection, Image Segmentation

Deployment: FastAPI, WebSocket, Docker, PostgreSQL, React

Tools: Git, Jupyter, Streamlit